

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)	(i)		Both distances / 20 cm and 1.5 m / <u>drop</u> height	1			1		1
		(ii)		Reduces effect of anomalies or random errors / identify anomalies (1) Accept outliers. Don't accept no anomalies So the <u>mean</u> is more <u>accurate</u> (1)	2			2		2
		(iii)		There are variations in the mass of each case (1) So weighing gives a more accurate value or results (1) OR Variations in the mass of each case is very small (1) Using the mean will not affect the results (1)		2		2		2
	(b)	(i)		As case <u>accelerates</u> the air resistance increases (1) until air resistance balances the weight of the case[s] / until the resultant force is zero / forces become balanced (1) Don't accept forces are equal or gravity	2			2		2
		(ii)		Measure split times e.g. over each 50 cm or 75 cm (1) All [split] times should be equal (1) OR Drop from >20 cm above pointer / drop from higher point (1) Check whether it affects the times / check the results are the same (1) OR Use multi-flash photographs / strobe (1) Check whether [successive] distances are equal (1)			2	2		2